The Hardware Barrier to Over–the–Air Adversarial RF–Fingerprint Spoofing: Why External Radiation Fails but Transmitter Pre–Distortion Succeeds

Hakim Lado Nirnimesh Nghose

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Abstract

Radio-frequency device-fingerprinting classifiers identify a transmitter from the hardware impairments—power-amplifier nonlinearity, I/Q imbalance, carrier frequency offset, and phase noise—stamped on its emissions. We ask whether a digital adversarial perturbation can spoof such a classifier over the air, and give a structural answer that turns on *whose hardware the perturbation passes through.* A targeted perturbation that flips the classifier from a legitimate device to a chosen target with probability one at the receiver input yields ≈0% fooling when it is radiated by a *second, external* transmitter: the classifier reads the emitter's hardware parameters, but every signal the adversary radiates is passed through the adversary's own—foreign and non-invertible—transmit operator, so it carries the *adversary's* fingerprint, not the target's. This barrier is strictly deeper than a channel-limited attack— expectation-over-transformation (channel-aware) crafting neutralizes the propagation channel but not the adversary's hardware operator, and over-the-air fooling stays ≈0%. By contrast, when the *same* perturbation is pre-distorted into the legitimate transmitter's own baseband—riding through its *invertible* hardware operator—the target *is* reachable: the required drive exists in closed form, and experimentally the classifier reads the target for 71% of frames at unit perturbation-to-signal ratio (versus 0% for a random perturbation of equal power, and 0% for the external adversary at every power). The hardware barrier therefore stops an external radiator but not pre-distortion at the legitimate transmitter—the decisive distinction being *addition through a foreign non-invertible operator* versus *pre-composition with an invertible own operator.*

We formalize both halves of this result: first (§5) that a targeted perturbation δ *does* reach the target when pre–distorted through the legitimate transmitter's own *invertible* hardware, with the required drive available in closed form; then that the same δ yields ≈ 0 fooling when instead radiated by a *second* transmitter, through that radio's foreign, *non–invertible* hardware (the hardware barrier, strictly deeper than Kim's channel–limited result).

Notation

Signals and hardware	
$n,\ u[n]$	sample index; intended baseband samples (digital content)
$\mathcal{T}_d$	transmit (hardware) operator of device d, eq. (1)
$g_d, M_d, c_d, \omega_d, \xi_d$	PA nonlinearity, I/Q imbalance, DC offset, CFO, phase noise of d
$\boldsymbol{\theta}_d$	fingerprint: impairment vector $(g_d, M_d, c_d, \omega_d, \xi_d)$
$h_d,\ w,\ y$	channel $d \to$RX; additive noise; received signal, eq. (3)
Classifier	
f	fingerprint classifier
$s_k(y)$	score (logit) for device k; decision $\hat{d}(y) = \arg\max_k s_k$
$S_k,\ \hat{\boldsymbol{\theta}}(y)$	score vs. impairment params; signature extracted from y
Attack	
x	captured device 6 frame ($f(x)$ =device 6); attack start point
$\delta,\ \delta^\star$	perturbation; PGD (digital) optimum
δ'	digital drive loaded into the adversary's DAC
$\varepsilon,\ \boldsymbol{e}_4,\ \mathrm{CE}$	budget $\\|\delta\\| \le \varepsilon \\|x\\|$; target one-hot (device 4); cross-entropy
device 6 / device 4	legit transmitter / spoofing target
Single-channel (pre-distortion)	
$\ell,\ \tau$	generic legitimate transmitter / target device (expt: dev 6 / dev 4)
$y_{\mathrm{SC}},\ J_\ell(u)$	single-channel emission eq. (8); Jacobian of $\mathcal{T}_\ell$ at u, eq. (11)
$\mathcal{P}_\ell$	emittable perturbation set through own hardware (full rank)
Over-the-air and EOT	
subscript a	adversary radio: $\mathcal{T}_a$, h_a, $\boldsymbol{\theta}_a$
$y_{\mathrm{OTA}},\ \mathcal{P}_a$	OTA composite eq. (15); physically radiable set
PSR	perturbation-to-signal ratio (dB)
$t,\ \mathcal{T}$	a transformation and its distribution (EOT)
$h,\ \mathcal{H}$	a channel realization and its distribution (EOT)
$\delta^\star_{\mathrm{EOT}}$	EOT (distribution-robust) perturbation

1 Hardware transmit operator (the fingerprint)

Each device d maps intended baseband samples $u[n] \in \mathbb{C}$ to an emitted RF baseband signal through a nonlinear transmit operator $\mathcal{T}_d$:

$$\mathcal{T}_d(u)[n] \;=\; g_d\big(M_d\, u[n] + c_d\big)\, e^{j\omega_d n} \;+\; \xi_d[n], \tag{1}$$

with power–amplifier nonlinearity $g_d(\cdot)$ (AM/AM + AM/PM, *non-invertible past saturation*), I/Q imbalance M_d, DC offset c_d, carrier frequency offset ω_d, and phase noise ξ_d. Collect the impairment parameters as the *fingerprint*

$$\boldsymbol{\theta}_d \;=\; (g_d,\, M_d,\, c_d,\, \omega_d,\, \xi_d). \tag{2}$$

2 Received signal and classifier

Through channel h_d with noise w,

$$y \;=\; h_d * \mathcal{T}_d(u) + w. \tag{3}$$

The classifier f is trained (per–window unit–RMS normalization; CFO/phase/time augmentation) to be invariant to content u and channel h and to read the impairments. Hence its class scores factor through an extracted signature $\hat{\boldsymbol{\theta}}(y)$:

$$s_k(y) \;\approx\; S_k\big(\hat{\boldsymbol{\theta}}(y)\big), \qquad \hat{d}(y) \;=\; \arg\max_k s_k(y). \tag{4}$$

Key: f is a function of the hardware parameters $\boldsymbol{\theta}$, not of the raw content.

3 Digital targeted PGD (construction of δ)

Take a captured device 6 frame x with $f(x) = \text{dev } 6$ (i.e. $\hat{\boldsymbol{\theta}}(x) \approx \boldsymbol{\theta}_6$). Solve, over an ℓ_2 ball of radius $\varepsilon\|x\|$,

$$\delta^\star \;=\; \underset{\|\delta\|\le\varepsilon\|x\|}{\arg\min}\; \mathrm{CE}\big(f(x+\delta),\, \boldsymbol{e}_4\big). \tag{5}$$

PGD ascends $\nabla_\delta s_4(x+\delta)$ *through the classifier*, so by construction

$$\delta^\star \in \text{input (signal) space of } f, \text{ defined by } \nabla f, \quad \text{such that } S_4(\hat{\boldsymbol{\theta}}(x+\delta^\star)) > S_6(\hat{\boldsymbol{\theta}}(x+\delta^\star)). \tag{6}$$

Empirically $f(x+\delta^\star) = \text{dev } 4$ with probability 1.

4 Digital evaluation succeeds

Digitally, $x+\delta^\star$ is fed *directly* to f; nothing sits between $\delta^\star$ and the input:

$$f\big(x+\delta^\star\big) = \text{dev } 4. \qquad \checkmark \tag{7}$$

5 Single–channel pre–distortion: the target *is* reachable

The attacker need not radiate δ from a separate device. Suppose the attacker instead controls the *digital baseband* fed to the legitimate transmitter and injects the perturbation there, *upstream* of the hardware: $u \mapsto u+\delta$. Write ℓ for the legitimate device and τ for the target (in our experiments $\ell = $ device 6, $\tau = $ device 4). The transmitted, then received, signal is

$$y_{\mathrm{SC}} \;=\; h_\ell * \mathcal{T}_\ell(u+\delta) \;+\; w. \tag{8}$$

We prove constructively that a budget–limited δ exists for which $f(y_{\mathrm{SC}}) = \tau$.

Step 1 — isolate what the attacker controls. The classifier reads the emitted signal, so what matters is how the emission moves when the baseband is perturbed. Split the emission into its clean part plus a controllable term,

$$\mathcal{T}_\ell(u+\delta) \;=\; \mathcal{T}_\ell(u) \;+\; e(\delta), \qquad e(\delta) \;:=\; \mathcal{T}_\ell(u+\delta) - \mathcal{T}_\ell(u). \tag{9}$$

At $\delta = 0$ the emission is the clean device–ℓ signal $\mathcal{T}_\ell(u)$, whose signature is $\boldsymbol{\theta}_\ell$, so $f = \ell$. The attacker steers the reading *only* through the emission perturbation $e(\delta)$; the whole question is which e are achievable.

Step 2 — linearize the hardware. For a budget–limited drive, expand $\mathcal{T}_\ell$ to first order about u:

$$e(\delta) \;=\; \mathcal{T}_\ell(u+\delta) - \mathcal{T}_\ell(u) \;=\; J_\ell(u)\,\delta \;+\; O(\|\delta\|^2), \qquad J_\ell(u) \;:=\; \left.\frac{\partial \mathcal{T}_\ell}{\partial u}\right|_u. \tag{10}$$

Differentiating the operator (1) sample–by–sample — the DC offset c_ℓ and phase noise ξ_ℓ do not depend on δ, the CFO $e^{j\omega_\ell n}$ is a fixed multiplier, the I/Q mixing M_ℓ is linear, and the PA g_ℓ acts pointwise — gives the Jacobian in closed form, a product of three per–sample factors,

$$J_\ell(u) \;=\; \underbrace{\mathrm{diag}\big(g'_\ell(M_\ell u[n] + c_\ell)\big)}_{\text{PA slope}} \cdot \underbrace{M_\ell}_{\text{I/Q mixing}} \cdot \underbrace{e^{j\omega_\ell n}}_{\text{CFO}}. \tag{11}$$

Step 3 — the Jacobian is invertible. Each factor of (11) is invertible under the radio's normal operating conditions:

- $\mathrm{diag}\big(g'_\ell(\cdot)\big)$ — invertible iff every entry $g'_\ell \neq 0$, i.e. the PA is driven *below hard saturation*, where its AM/AM curve is strictly increasing;
- M_ℓ — invertible iff $\det M_\ell \neq 0$, true for any physical front end, on which the I/Q imbalance is a small perturbation of the identity;
- $e^{j\omega_\ell n}$ — unit modulus, hence invertible at every sample.

A product of invertible maps is invertible, so $J_\ell(u)$ has *full rank*.

Step 4 — the reachable set is full–dimensional. The emission perturbations the attacker can produce within the budget $\|\delta\| \leq \varepsilon\|u\|$ form

$$\mathcal{P}_\ell \;=\; \big\{\, e(\delta) : \|\delta\| \leq \varepsilon\|u\| \,\big\} \;\approx\; J_\ell(u)\cdot \mathrm{Ball}\big(\varepsilon\|u\|\big). \tag{12}$$

A full–rank linear map carries a ball to a full–dimensional ellipsoid about the origin. Hence $\mathcal{P}_\ell$ contains a neighbourhood of 0 in emission space and *grows to cover every direction* as ε increases: no emission direction is blocked.

Step 5 — reaching the target, in closed form. By construction (§3) the digital optimum $\delta^\star_{\mathrm{dig}}$ shifts the extracted signature into the τ–decision region. Because f is channel–invariant (§2), this shift is a property of the *emission*: let $e^\star$ be the emission–domain perturbation that carries the signature to τ. Reaching the target then means realizing $e(\delta) = e^\star$, which by Step 3 has the closed–form solution

$$\boxed{\delta^\star \;\approx\; J_\ell(u)^{-1}\, e^\star} \tag{13}$$

The drive $\delta^\star$ *exists*: the invertible own–hardware Jacobian synthesizes the required emission perturbation from a baseband drive, and by Step 4 $e^\star \in \mathcal{P}_\ell$ once $\varepsilon \geq \|\delta^\star\|/\|u\|$.

Step 6 — the crossover exists. As ε grows from 0, the reading moves continuously from device ℓ along the reachable direction $e^\star$. Since $e^\star$ is, by definition, the direction the classifier reads as τ, and it lies in $\mathcal{P}_\ell$ for ε large enough (Step 4), there is a finite drive at which the decision crosses into the target:

$$\boxed{\exists\, \varepsilon^\star : \quad f\big(h_\ell * \mathcal{T}_\ell(u+\delta^\star)\big) = \tau, \qquad \|\delta^\star\| = \varepsilon^\star\|u\|.} \tag{14}$$

The mechanism is that δ is *pre–composed* with the transmitter's own hardware: it rides through the very operator $\mathcal{T}_\ell$ that stamps the legitimate fingerprint, and because that operator's Jacobian is full rank, the hardware passes the perturbation faithfully into every emission direction rather than collapsing it. □

Experimental confirmation. With correctly scaled δ (realized PSR = label) injected into the legitimate transmitter's baseband, the reading flips to the target at the $\varepsilon^\star$ corresponding to PSR $\approx$ 0 dB ($\|\delta\| \approx \|u_{\mathrm{payload}}\|$): $f(y_{\mathrm{SC}}) = \tau$ for 71% of frames, versus 0% for a *random* δ of equal power — confirming the flip follows the crafted direction $e^\star$ and is not loud–signal collapse.

6 Over-the-air (two-channel) signal

Device 6 emits the legit frame $h_6 * \mathcal{T}_6(u)$ (carries $\boldsymbol{\theta}_6$); the adversary loads a digital drive δ' into *its* DAC and emits $h_a * \mathcal{T}_a(\delta')$ (carries $\boldsymbol{\theta}_a$). They superimpose linearly in the air:

$$y_{\mathrm{OTA}} \;=\; h_6 * \mathcal{T}_6(u) \;+\; h_a * \mathcal{T}_a(\delta') \;+\; w. \tag{15}$$

We require $f(y_{\mathrm{OTA}}) = \text{dev } 4$.

7 The exact condition, and why it cannot hold

To reproduce the digital success the radiated perturbation must equal $\delta^\star$:

$$h_a * \mathcal{T}_a(\delta') \overset{!}{=} \delta^\star \quad\Longrightarrow\quad \delta' \;=\; \mathcal{T}_a^{-1}\Big(h_a^{-1}\big(\delta^\star\big)\Big). \tag{16}$$

This is infeasible: (i) $\mathcal{T}_a^{-1}$ does not exist (g_a non-invertible past saturation; $\delta^\star$ may lie outside range($\mathcal{T}_a$)); (ii) h_a is unknown without channel awareness; (iii) every physically radiated signal is passed through $\mathcal{T}_a$, which *imprints* $\boldsymbol{\theta}_a$ *on everything it emits.*

8 Reachable-set argument (core result)

Define the set of perturbations the adversary can physically place on the air:

$$\mathcal{P}_a \;=\; \big\{\, h_a * \mathcal{T}_a(\delta') \;:\; \|\delta'\| \le \varepsilon \,\big\}, \qquad \text{every element carries } \boldsymbol{\theta}_a. \tag{17}$$

The OTA attack succeeds iff $\exists\, p \in \mathcal{P}_a$ with $f\big(h_6 * \mathcal{T}_6(u) + p\big) = \text{dev } 4$. Because f reads impairments and every $p \in \mathcal{P}_a$ carries $\boldsymbol{\theta}_a$, the composite signature is a mixture along the $\boldsymbol{\theta}_6$–$\boldsymbol{\theta}_a$ axis parameterized by the perturbation–to–signal ratio (PSR):

$$\hat{\boldsymbol{\theta}}\big(h_6 * \mathcal{T}_6(u) + p\big) \;=\; \Theta\big(\boldsymbol{\theta}_6, \boldsymbol{\theta}_a;\, \text{PSR}\big). \tag{18}$$

As PSR grows, the reading slides dev 6 $\to$ (device with fingerprint $\approx \boldsymbol{\theta}_a$), and

$$\boxed{f(y_{\mathrm{OTA}}) = \text{dev } 4 \quad\Longleftrightarrow\quad \boldsymbol{\theta}_a \approx \boldsymbol{\theta}_4} \tag{19}$$

i.e. only if the adversary radio *is* device 4. The digital solution $\delta^\star$ is a direction in ∇f–space; $\mathcal{P}_a$ is a manifold in hardware–$\boldsymbol{\theta}_a$–space; generically $\delta^\star \notin \mathcal{P}_a$. This is the observed 0%.

This is the mirror image of the single–channel case (§5): there δ was *pre–composed* with the transmitter's *own* invertible operator $\mathcal{T}_\ell$, so the required drive existed in closed form (13); here δ' is instead *added through a foreign* operator $\mathcal{T}_a$ that stamps $\boldsymbol{\theta}_a$ on everything it emits and admits no such inverse. The decisive distinction is *pre–composition with an invertible own operator* versus *addition through a foreign non–invertible one*:

$$\underbrace{\mathcal{T}_\ell(u + \delta)}_{\delta \textit{ inside } \mathcal{T}_\ell \text{ (invertible)}} \qquad \text{vs.} \qquad \underbrace{\mathcal{T}_\ell(u) + h_a * \mathcal{T}_a(\delta')}_{\delta' \textit{ through} \text{ foreign } \mathcal{T}_a}. \tag{20}$$

9 Contrast with Kim (channel–limited attack)

If instead the classifier reads *content* (modulation), $s_k(y) \approx S_k(c(y))$, the perturbation acts on content, which traverses the linear part and is channel–correctable. Making δ channel–aware [2] handles h_a^{-1} and the perturbation reaches the content decision region: the attack *works*. There the sole barrier is the channel h_a. Here the barrier is the hardware operator $\mathcal{T}_a$, which no digital drive from a foreign radio can invert, because the target $\boldsymbol{\theta}_4$ is a property of the *emitter*, not of the injected signal.

10 Prediction for Expectation–over–Transformation (EOT)

EOT [1] robustifies over the channel,

$$\delta^{\star}_{\mathrm{EOT}} \;=\; \arg\max_{\|\delta\| \le \varepsilon \|x\|} \; \mathbb{E}_{h \sim \mathcal{H}}\big[\, s_4\big(h(x+\delta)\big)\,\big], \tag{21}$$

which neutralizes h_a—the whole of Kim's barrier.[1] But EOT does nothing to $\mathcal{T}_a$: the emitted signal is still $\mathcal{T}_a(\delta'_{\mathrm{EOT}})$ and still carries $\boldsymbol{\theta}_a$. Therefore

$$\text{Kim's barrier } (h_a): \text{ closed by EOT;} \qquad \text{hardware barrier } (\mathcal{T}_a): \text{ untouched by EOT.} \tag{22}$$

Prediction (confirmed in §11): EOT remains $\approx 0\%$ toward device 4 unless $\boldsymbol{\theta}_a \approx \boldsymbol{\theta}_4$. If the EOT run still fails, the barrier is hardware ($\mathcal{T}_a$), strictly deeper than Kim's channel result; if it succeeds, the limitation was only the channel, i.e. the same as Kim.

11 Experimental confirmation

Session-14 over-the-air measurements (hardware-fingerprint CVNN; device 6 legit, device 4 target; targeted PGD; δ confined to the payload; pristine $\mathcal{T}_6(u)$ on ch0 so device 6 stamps its fingerprint once). The classifier scoring the captures is domain-adapted to the replay so the δ-off baseline reads device 6.

Setting	targeted → device 4
Digital (δ at receiver input)	**100%**
OTA, no-EOT (2-channel)	$\approx 0\%$ (all PSR)
OTA, channel-aware EOT (2-channel)	$\approx 0\%$ (all PSR)

Per-PSR fraction (channel-aware EOT sweep, replay-adapted model):

PSR (dB)	→ device 4	→ device 6
0 (loudest δ)	0.00	1.00
-5	0.04	0.75
-10	0.00	0.96
-15	0.04	0.95
-20	0.01	0.96
-30 ($\approx \delta$-off)	$0.21^{\dagger}$	0.68

[1] EOT (Expectation over Transformation) is *not* channel estimation. Channel estimation *measures* the specific channel h_a and inverts it ($\delta' = h_a^{-1}(\delta^\star)$); it needs pilots/feedback on that link. EOT instead maximizes the *expected* objective over a modeled *distribution* of channel realizations $\mathcal{H}$ (here random sub-sample time-shift and carrier-phase rotation), producing a δ robust to the whole family *without* estimating any single channel. We use EOT because the adversary→receiver link carries no pilot/feedback, so h_a is unobservable. Both, however, only address h_a; neither touches $\mathcal{T}_a$.

† residual baseline mis-read noise, *anti-correlated* with δ power (largest where δ is weakest); a thr=2 extraction merged the near-δ-off capture into 16 s bursts and spuriously reported 0.60, corrected to 0.21 with thr ≥ 4.

At the loudest perturbation (PSR 0 dB) the composite still reads device 6 with probability 1.00, and the fooling rate toward device 4 does *not* increase with perturbation power. Both the naive and the channel-aware over-the-air attacks give $\approx 0\%$ toward device 4. This confirms the prediction: EOT neutralizes the channel h_a but not the hardware operator $\mathcal{T}_a$, so the barrier is hardware — strictly deeper than Kim's channel-limited result.

References

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[2] B. Kim, Y. E. Sagduyu, K. Davaslioglu, T. Erpek, and S. Ulukus, "Channel-Aware Adversarial Attacks Against Deep Learning-Based Wireless Signal Classifiers," *IEEE Trans. Wireless Commun.*, vol. 21, no. 6, pp. 3868–3880, 2022.